F12 4th CAR

Lecture 30: Section 4.7
Inverse Trigonometric Functions

Inverse sine function

Inverse cosine function

Inverse tangent function

Inverse properties

Review: Function - each x has only one y, passes vertical line test.

Review: One-to-one - each y has only one x, passes horizontal line test.

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Reminder: The function and its inverse must must be functions: each x has only one y, passes the vertical line test.

Lecture 30: Section 4.7

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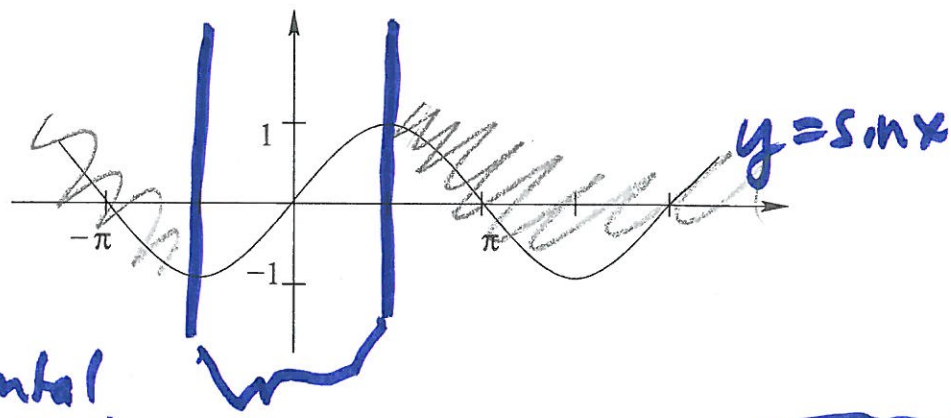
Inverse properties

$\frac{1}{2} = \sin \frac{\pi}{6}$

Also, must be one-to-one: each y has only one x, passes the horizontal line test.

Lecture 30: Section 4.7
Inverse Trigonometric Functions

The Inverse Sine Function



To pass horizontal line test

Let's restrict the domain to the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. *

Then $y = \sin x$ is one-to-one.

R: $[-1, 1]$

<u>Def.</u> The **inverse sine function** is defined by

$$y = \sin^{-1} x \quad \text{if and only if} \quad x = \sin y$$

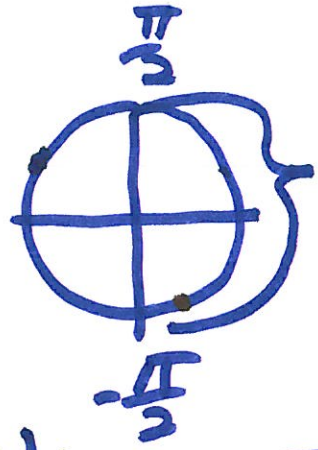
with domain $[-1, 1]$ and range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

NOTE: The inverse sine function is also called **arcsine**, denoted by $y = \arcsin x$. $= \sin^{-1} x$

~~Remind~~

Reminder: $f^{-1}(x) \neq \frac{1}{f(x)} = [f(x)]^{-1}$

thus $\sin^{-1} x \neq \frac{1}{\sin x} = (\sin x)^{-1}$



ex. Find the exact value, if possible.

$\frac{\pi}{6}$ 1) $\sin^{-1}\left(\frac{1}{2}\right) = y$

$\sin y = \frac{1}{2}$, thus $y = \frac{\pi}{6}$ and $-\frac{\pi}{2} \le \frac{\pi}{6} \le \frac{\pi}{2}$

$-\frac{\pi}{3}$ 2) $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = y$

$\sin y = -\frac{\sqrt{3}}{2}$, thus $y = -\frac{\pi}{3}$ and $-\frac{\pi}{2} \le -\frac{\pi}{3} \le \frac{\pi}{2}$

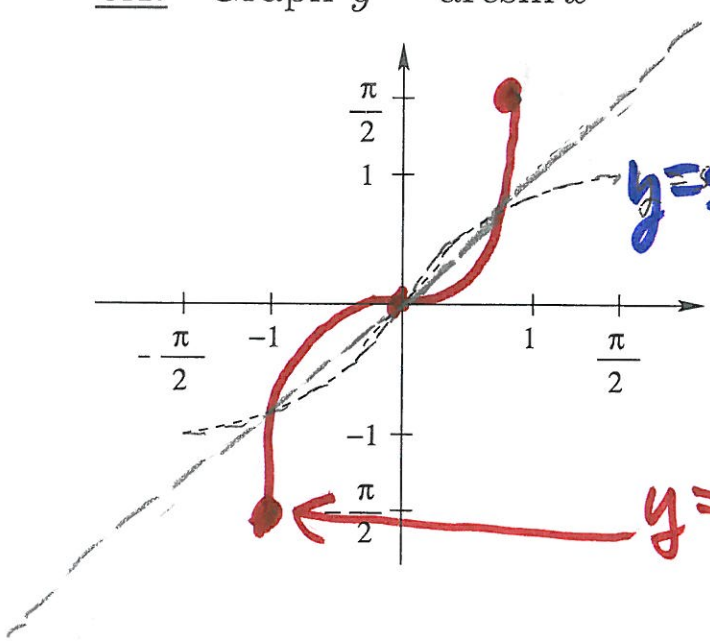
does not exist 3) $\sin^{-1}\left(\frac{3}{2}\right) = y$

$\sin y = \frac{3}{2}$

not possible

$-1 \le \sin\theta \le 1$

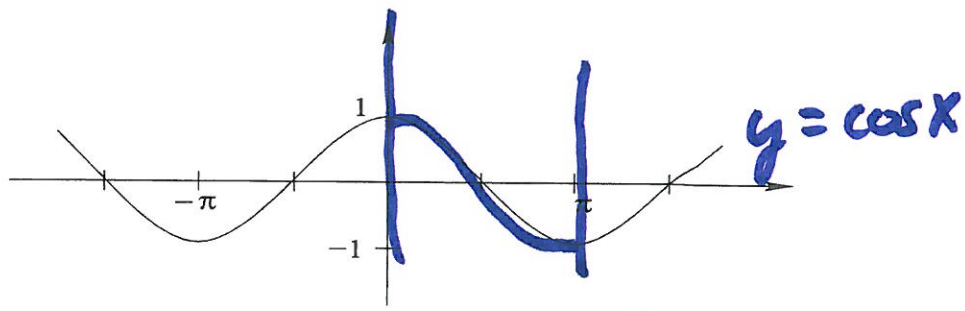
ex. Graph $y = \arcsin x$



x	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
y	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1

x	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1
y	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$

The Inverse Cosine Function



Restrict the domain to the interval $[0, \pi]$: — to get a 1-1 function

<u>Def.</u> The **inverse cosine function** (or **arccosin function**, denoted by $y = \arccos x$) is defined by

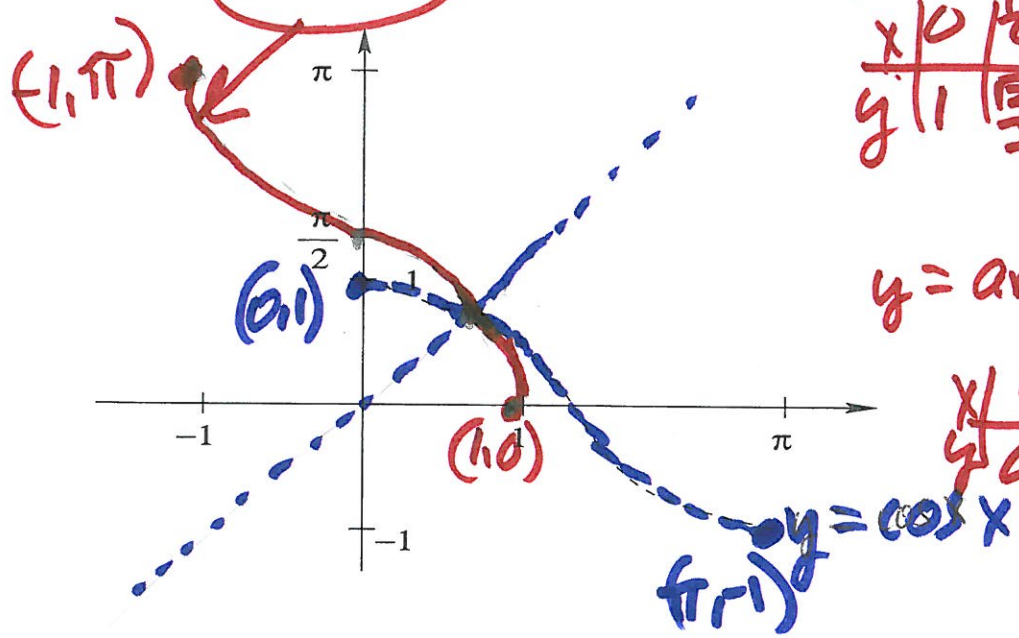
angle → $y = \cos^{-1} x$ if and only if $x = \cos y$

with domain $[-1, 1]$ and range $[0, \pi]$.

Graph $y = \arccos x$

$y = \cos x$

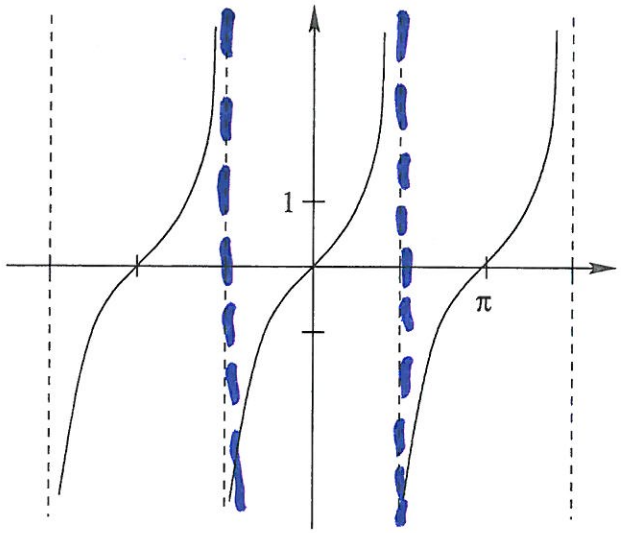
x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1

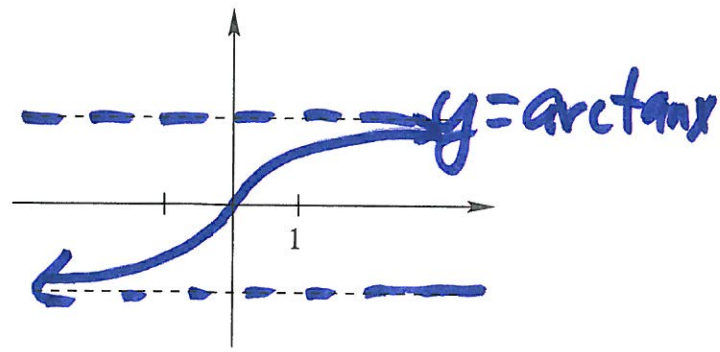


$y = \arccos x$

x	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1
y	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π

The Inverse Tangent Function





Restrict the domain to the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

<u>Def.</u> The **inverse tangent function** (or **arctangent function**, denoted by $y = \arctan x$) is defined by

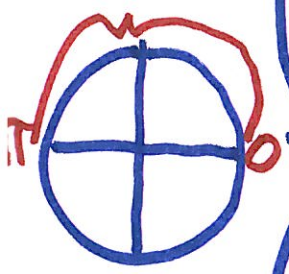
angle

$$y = \tan^{-1} x \quad \text{if and only if} \quad x = \tan y$$

with domain $(-\infty, \infty)$ and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

NOTE: The graph of $y = \tan^{-1} x$ has two horizontal asymptotes:

ex. Find



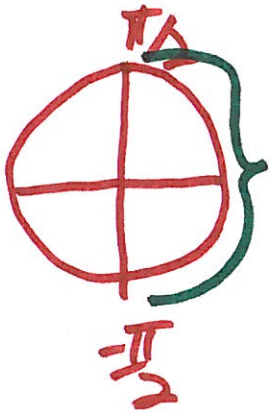
1) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = y$ $\quad \cos y = \frac{\sqrt{3}}{2}$, $y = \frac{\pi}{6}$ and $0 \le \frac{\pi}{6} \le \pi$

$= \frac{\pi}{6}$

2) $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

3) $\arccos(0) = \frac{\pi}{2}$ since $\cos\frac{\pi}{2} = 0$ and $0 \le \frac{\pi}{2} \le \pi$

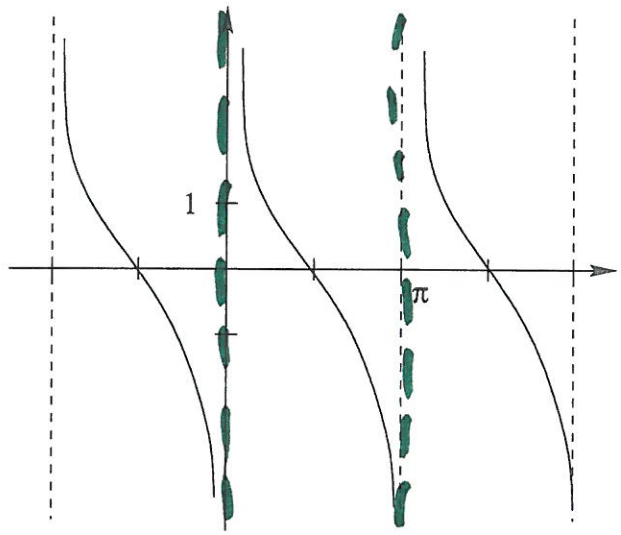
4) $\arctan(0) = 0$ since $\tan 0 = 0$ and $-\frac{\pi}{2} < 0 < \frac{\pi}{2}$

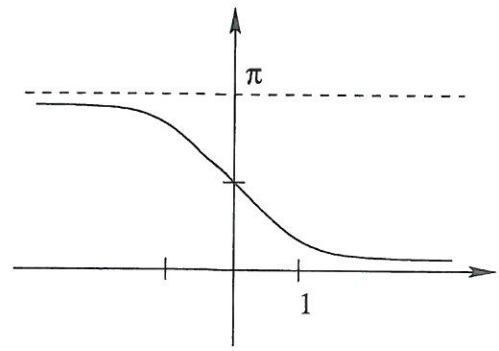


5) $\arctan(1) = \frac{\pi}{4}$

6) $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

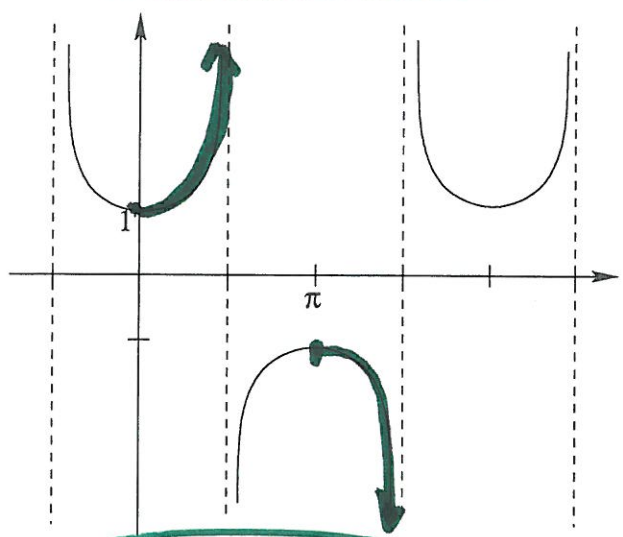
Other Inverse Trigonometric Functions

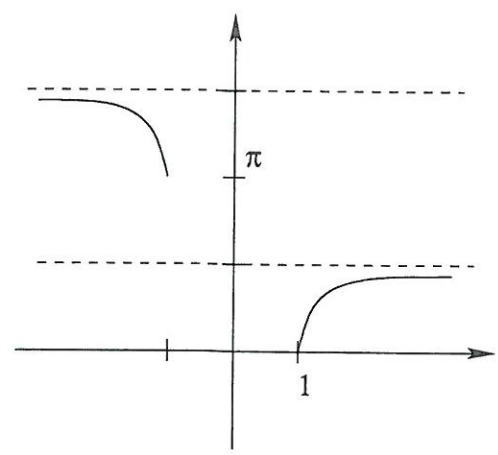




$y = \cot x,\ 0 < x < \pi$

$y = \cot^{-1} x$

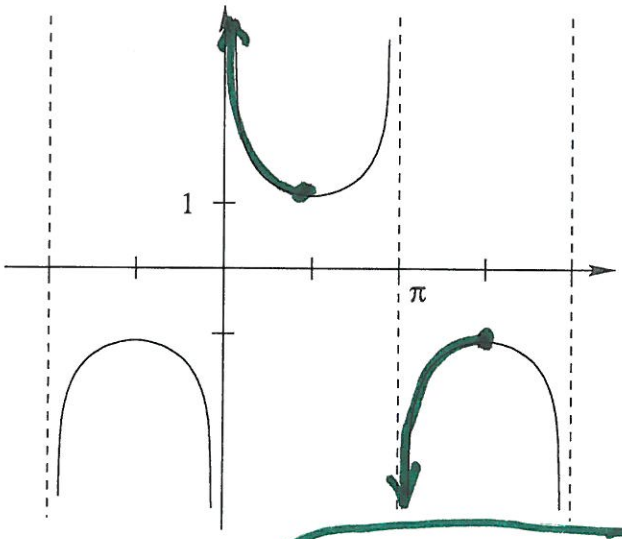




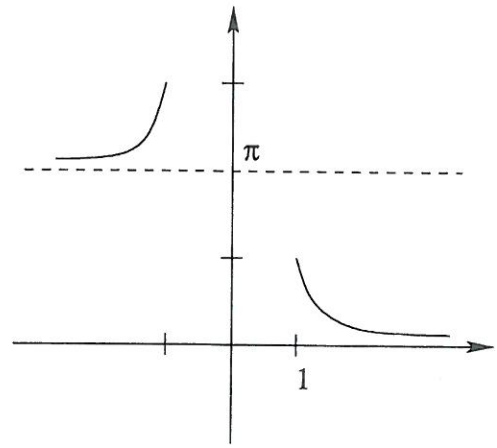
$y = \sec x,\ 0 \le x < \frac{\pi}{2},\ \pi \le x < \frac{3\pi}{2}$

$y = \sec^{-1} x$

$$f(f^{-1}(x)) = x = f^{-1}(f(x))$$



$y = \csc x, \; 0 < x \leq \frac{\pi}{2}, \; \pi < x \leq \frac{3\pi}{2}$



$y = \csc^{-1} x$

Inverse Properties Short-cuts (but can be tricky)

1. $\sin(\sin^{-1} x) = x \quad$ for $-1 \leq x \leq 1$

   $\sin^{-1}(\sin x) = x \quad$ for $-\dfrac{\pi}{2} \leq x \leq \dfrac{\pi}{2}$ tricky because of domain restriction

2. $\cos(\cos^{-1} x) = x \quad$ for $-1 \leq x \leq 1$

   $\cos^{-1}(\cos x) = x \quad$ for $0 \leq x \leq \pi$

3. $\tan(\tan^{-1} x) = x \quad$ for all x

   $\tan^{-1}(\tan x) = x \quad$ for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$

ex. Find the exact value, if it is possible.

1) $\arctan\left(\tan\frac{\pi}{6}\right) = \frac{\pi}{6}$ and $-\frac{\pi}{2} < \frac{\pi}{6} < \frac{\pi}{2}$

π is not in the domain.

2) $\sin(\arcsin \pi) =$ D.N.E No sines are bigger than one.

too big for short-cut

3) $\arcsin\left(\sin\frac{5\pi}{6}\right) =$

$\arcsin\left(\frac{1}{2}\right) = \boxed{\frac{\pi}{6}}$

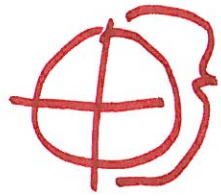
$-\frac{\pi}{2} \le \quad \le \frac{\pi}{2}$

4) $\cos^{-1}\left(\cos -\frac{\pi}{4}\right) = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

$= \boxed{\frac{\pi}{4}}$

$0 \le \frac{\pi}{4} \le \pi$

tricky since $-\frac{\pi}{4}$ is not in $[0, \pi]$

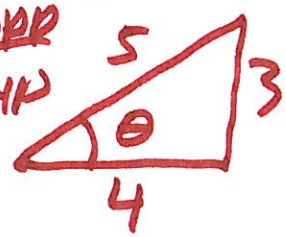


cosine is +

ex. Find the exact value:

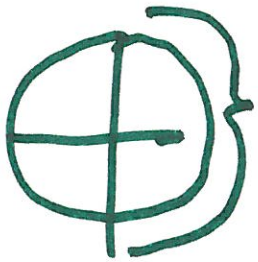
1) $\cos\left(\underbrace{\arcsin\frac{3}{5}}_{\theta}\right) = \cos\theta = \frac{adj}{hyp} = \boxed{\frac{4}{5}}$

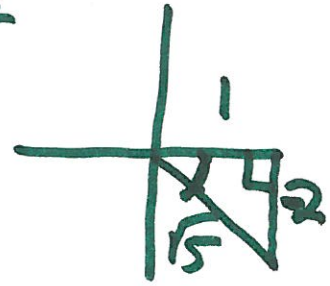
$\sin\theta = \frac{3}{5} = \frac{opp}{hyp}$



2) $\cos(\underbrace{\tan^{-1}(-2)}_{\theta}) = \cos\theta = \frac{adj}{hyp} = \frac{1}{\sqrt{5}}\left(\frac{\sqrt{5}}{\sqrt{5}}\right) = \frac{\sqrt{5}}{5}$

$\tan\theta = -2 = \frac{opp}{adj} = \frac{-2}{1}$





$1^2 + 2^2 = h^2$

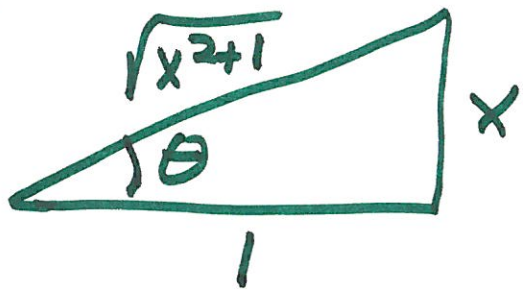
$5 = h^2$

$h = \sqrt{5}$

ex. Rewrite the expression as an algebraic expression:

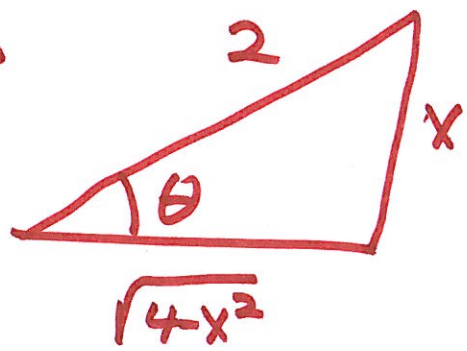
1) $\cos(\overbrace{\tan^{-1} x}^{\theta}) = \cos\theta = \frac{adj}{hyp} = \frac{1}{\sqrt{x^2+1}} = \frac{\sqrt{x^2+1}}{x^2+1}$

$\tan\theta = x = \frac{x}{1} = \frac{opp}{adj}$



2) $\sec\left(\underbrace{\arcsin\frac{x}{2}}_{\theta}\right) = \sec\theta = \frac{hyp}{adj} = \frac{2}{\sqrt{4-x^2}}$

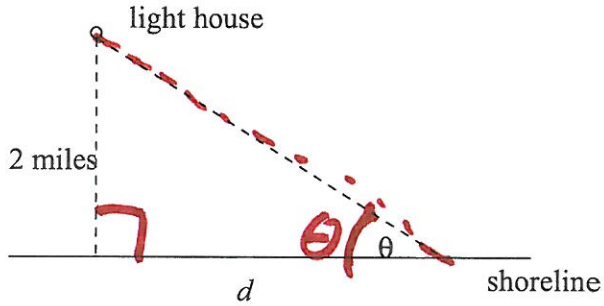
$\sin\theta = \frac{x}{2} = \frac{opp}{hyp}$



$= \frac{2\sqrt{4-x^2}}{4-x^2}$

Applications

<u>ex.</u> A light house is located on an island that is 2 miles off a straight shoreline.



1) Express the angle θ formed by the beam of light and the shoreline in terms of the distance d.

$\tan\theta = \frac{2}{d}$

$\theta = \arctan\frac{2}{d}$

2) Find θ when $d = 4$ miles.

$\theta = \arctan\frac{2}{4} = \arctan\frac{1}{2}$

need calculator

Practice.

Find the exact value:

1) $\sin^{-1}\left(\sin\frac{9\pi}{10}\right)$

2) $\cos^{-1}\left(\cos\frac{16\pi}{5}\right)$

Answer. 1) $\frac{\pi}{10}$ 2) $\frac{4\pi}{5}$